PayLens Fintech Platform

Project Proposal | Data Engineering Track

Project Overview

Project Name: PayLens Fintech Platform (Real-Time Merchant Intelligence, Risk & Settlement Platform)
Track: Data Engineering

Objective

To build an end-to-end data engineering platform engineered for Payment Service Providers (PSPs) and e-commerce merchants. The platform serves as a centralized intelligence engine uniting streaming payment events, batch master data, explainable risk detection models, dynamic financial settlement calculations, and 3-way automated reconciliation pipelines.

Problem Statement & Financial Bottlenecks

Modern digital payment infrastructures suffer from critical financial engineering bottlenecks that hinder operational efficiency and accurate reporting:

1. **Information Fragmentation & Data Disparity:** Real-time payment attempts stream at high velocity, whereas merchant master files, fee structures, and bank statement logs update via slow batch processing. Blending these disparate sources with low latency is highly complex.
2. **Risk vs. Conversion Dilemma:** Blocking customer transactions automatically upon fraud suspicion creates severe false-positive rates. Payment engines require non-disruptive Explainable Risk Rules triggering "REVIEW REQUIRED" alerts without stopping live funds.
3. **Settlement Complexity & Ledger Integrity:** Calculating exact net payouts requires navigating intricate financial lifecycles (sales, refunds, chargebacks with dispute fees, and rolling platform fee deductions) under append-only ledger rules.

4. **Reconciliation Gaps:** Matching internal transaction records against third-party processor logs and bank payout reports is traditionally manual, error-prone, and improperly aligned.

Technology Architecture & Tools

The platform leverages modern cloud data tooling structured across a robust Medallion Architecture:

Layer / Functional Area	Technology & Tools	Description / Responsibility
Fast / Streaming Ingestion	NestJS API, Azure Event Hubs	High-velocity streaming of live payment event data
Slow / Batch Ingestion	Azure Data Factory	Ingestion of merchant master data, fee schedules, and bank logs
Bronze Layer (Raw)	Azure Blob Storage	Immutable, landing storage for raw streaming and batch payloads
Silver Layer (Validated)	Delta Lake, PySpark	Deduplicated, cleaned, schema-enforced transactional data
Gold Layer (Analytics)	PostgreSQL, Azure Synapse	Multi-tenant dimensional models and optimized analytical star schemas
Platform & Versioning	Python, SQL, Git & GitHub	Core programming models, transformations, and repository workflows

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Team Members & Key Responsibilities

Team Member	Role & Assigned Focus	Core Responsibilities
Mostafa Hamdy Abdelrazek Abdelghany	Lead Pipeline Architect	Overall Pipeline Architecture, Core Orchestration & Workflow Management
Yomna Khalid Ali Ayoub	Ingestion Engineer	Streaming Ingestion (Event Hubs) & Batch Ingestion Pipelines (Azure Data Factory)
Alaa Hamdeen Mohammad Ghanem	Data Platform Engineer	Medallion Architecture (Bronze/Silver/Gold PySpark ETL) & Dimensional Data Modeling
Aya Hesham Saad Mahmoud	Financial Logic Engineer	Settlement Engine Logic, 3-Way Reconciliation Module & Multi-Tenant Analytics

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Execution Plan & Deliverables

Week 1: Requirements & Design ⟶ Week 2: Ingestion & Bronze ⟶ Week 3: Silver/Gold Processing ⟶ Week 4: Settlement & Final

Week 1: Project Planning & Requirements

- **Key Tasks:** Finalize project proposal, establish GitHub repository architecture and branch policies, define standardized data contracts across streaming and batch interfaces.
- **Deliverables:** Approved Proposal, System Architecture Blueprint, Data Contract & Requirements Specification.

Week 2: Ingestion & Bronze Layer Implementation

- **Key Tasks:** Configure NestJS API and Azure Event Hubs for continuous event streaming; build Azure Data Factory (ADF) pipelines for scheduled batch ingestion.
- **Deliverables:** Fully functional ingestion pipelines and append-only, immutable Bronze layer storage on Azure Blob Storage.

Week 3: Silver & Gold Transformation Layers

- **Key Tasks:** Implement PySpark Delta Lake `MERGE` statements for incremental processing, build automated deduplication and quality validation rules, build Gold star schema datasets in PostgreSQL and Azure Synapse.
- **Deliverables:** Validated Silver Delta tables, structured Gold dimensional schema for merchant reporting.

Week 4: Settlement Engine, Reconciliation & Final Rollout

- **Key Tasks:** Develop automated financial settlement calculation logic, construct the 3-way ledger reconciliation module, finalize multi-tenant analytics dashboards and reporting interfaces.
- **Deliverables:** Fully integrated end-to-end pipeline, automated 3-way reconciliation reporting, complete project documentation and final GitHub repository release.